

Cher Ming Tan

professor

基本信息

姓名: Cher Ming Tan

所属院校: 南洋理工大学

地区: Singapore

年龄范围: 50岁以上

海外经历: 有

企业经历: 有

荣誉称号

★ IEEE Fellow

这些成就代表了在学术领域的最高认可，展现了卓越的研究能力和影响力。

学科领域

材料物理

微电子科学与工程

材料科学与工程

集成电路设计与集成系统

纳米材料与技术

研究领域

微电子封装与可靠性

固体物理与材料力学性能

材料物理与化学

EDA工具与电路验证

材料微结构与表征

功率器件物理与结构

材料失效分析与可靠性

数字集成电路设计

纳米复合材料

薄膜材料与表面物理

+4

个人简介

Dr. Cher Ming Tan, a distinguished professor in the Electronic Department at Chang Gung University, Taiwan, holds a Ph.D. in Electrical Engineering from the

University of Toronto. With extensive experience in electronic industry reliability, he has authored over 300 international journal and conference papers, secured 12 patents, and written four books. Dr. Tan is an editor for Scientific Reports, IEEE TDMR, and SpringerBrief in Reliability, and a member of IEEE's advisory panel for Elsevier Publishing Group. He has received the IEEE Region 10 Outstanding Volunteer Award and is a Fellow of the Institute of Engineers, Singapore, and the Singapore Quality Institute. His research focuses on reliability and failure physics modeling of electronic systems. For more information, visit www.chemmingtan.com.

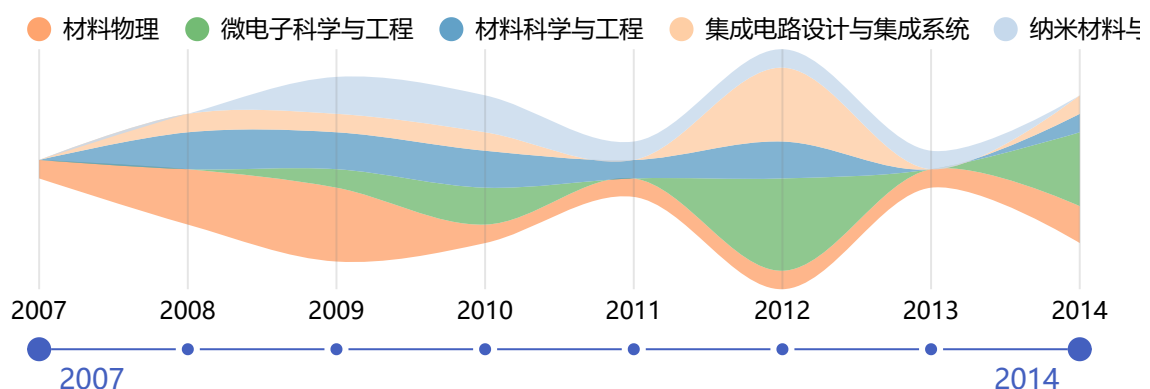
工作经历

LiteOn Power Semiconductor Corporation - Special Assistant	1991 - 1997
Hewlett-Packard Inc - R&D Engineer	1986 - 1987
Fairchild Semiconductor Corp - Failure analysis engineer	1984-05 - 1986-02

教育经历

University of Toronto - M.A. Sc and Ph.D	1986-09 - 1992-04
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近年发表的论文



Rapid ULSI Interconnect Reliability Analysis Using Neural Networks

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Tian, Yizhen; He, Feifei; Zhang, Qi-Jun; Tan, Cher Ming; Ma, Jianguo

被引用次数: 18

2014

Modeling and analysis of gate-all-around silicon nanowire FET

MICROELECTRONICS RELIABILITY

Chen, Xiangchen; Tan, Cher Ming

被引用次数: 32

2014

Maintenance Scheduling of Plasma Etching Chamber in Wafer Fabrication for High-Yield Etching Process

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Le Minh Duc; Tan, Cher Ming; Luo, Ming; Leng, Ian Chan Hian

被引用次数: 13

2014

Time Evolution Degradation Physics in High Power White LEDs Under High Temperature-Humidity Conditions

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Tan, Cher Ming; Singh, Preetpal

被引用次数: 49

2014

A practical framework of electrical based online state-of-charge estimation of lithium ion batteries

JOURNAL OF POWER SOURCES

Leng, Feng; Tan, Cher Ming; Yazami, Rachid; Minh Duc Le

被引用次数: 54

2014

Methodology of reliability enhancement for high power LED driver

MICROELECTRONICS RELIABILITY

Lan, Song; Tan, Cher Ming; Wu, Kevin

被引用次数: 12

2014

Special section reliability and variability of devices for circuits and systems

MICROELECTRONICS RELIABILITY

Asenov, Asen; Schlichtmann, Ulf; Tan, Cher Ming; Wong, Hei; Zhou, Xing

被引用次数: 0

2014

Damage threshold determination and non-destructive identification of possible failure sites in PIN limiter

MICROELECTRONICS RELIABILITY

Tan, Cher Ming; Yu, Wen Zhi

被引用次数: 1

2014

Ab initio simulation of electronic and mechanical properties of aluminium for fatigue early feature investigation

INTERNATIONAL JOURNAL OF NANOTECHNOLOGY

Zhang, Shuai; Tan, Cher Ming; Cheng, Shuguang; Deng, Tianqi; He, Feifei; Su, Haibin

被引用次数: 3

2014

Effects of Carbon Loading on the Performance of Functionalized Carbon Nanotube Polymer Heat Sink for High Power Light-Emitting Diode in Switching Applications

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Tan, Cher Ming; Chen, Sihan Joseph; Kong, Jacky

被引用次数: 2

2013

Optimal maintenance strategy of deteriorating system under imperfect maintenance and inspection using mixed inspection scheduling

RELIABILITY ENGINEERING & SYSTEM SAFETY

Minh Duc Le; Tan, Cher Ming

被引用次数: 79

2013

Systematic Root Cause Analysis for GaP Green Light LED Degradation

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Tan, Cher Ming; Lai, Chao Sung

被引用次数: 1

2013

Applications of multi-walled carbon nanotube in electronic packaging

NANOSCALE RESEARCH LETTERS

Tan, Cher Ming; Baudot, Charles; Han, Yongdian; Jing, Hongyang

被引用次数: 15

2012

Ensuring accuracy in optical and electrical measurement of ultra-bright LEDs during reliability test

MICROELECTRONICS RELIABILITY

Chen, Sihan Joseph; Tan, Cher Ming; Chen, Boon Khai Eric; Chua, Zhi Yong Shaun

被引用次数: 3

2012

ICMAT 2011-Reliability and variability of semiconductor devices and ICs

MICROELECTRONICS RELIABILITY

Asenov, Asen; Schlichtmann, Ulf; Tan, Cher Ming; Wong, Hei; Zhou, Xing

被引用次数: 0

2012

Effect of IC layout on the reliability of CMOS amplifiers

MICROELECTRONICS RELIABILITY

He, Feifei; Tan, Cher Ming

被引用次数: 2

2012

Applications of finite element methods for reliability study of ULSI interconnections

MICROELECTRONICS RELIABILITY

Tan, Cher Ming; Li, Wei; Gan, Zhenghao

被引用次数: 10

2012

Reliability study of LED driver - A case study of black box testing

MICROELECTRONICS RELIABILITY

Lan, Song; Tan, Cher Ming; Wu, Kevin

被引用次数: 23

2012

Rapid Light Output Degradation of GaN-Based Packaged LED in the Early Stage of Humidity Test

IEEE TRANSACTIONS ON DEVICE AND MATERIALS RELIABILITY

Tan, Cher Ming; Chen, B. K.; Li, X.; Chen, Sihan Joseph

被引用次数: 22

2012

Electromigration reliability of interconnections in RF low noise amplifier circuit

MICROELECTRONICS RELIABILITY

He, Feifei; Tan, Cher Ming

被引用次数: 13

2012

Comparison of electromigration simulation in test structure and actual circuit

APPLIED MATHEMATICAL MODELLING

He, Feifei; Tan, Cher Ming

被引用次数: 1

2012

Integration of Low- κ Dielectric Liner in Through Silicon Via and Thermomechanical Stress Relief

APPLIED PHYSICS EXPRESS

Ghosh, Kaushik; Zhang, Jiye; Zhang, Lin; Dong, Yuanwei; Li, Hongyu; Tan, Cher Ming; Xia, Guangrui; Tan, Chuan Seng

被引用次数: 11

2012

Electrical-Thermal-Stress Coupled-Field Effect in SOI and Partial SOI Lateral Power Diode

IEEE TRANSACTIONS ON POWER ELECTRONICS

Huang, Guang Yu; Tan, Cher Ming

被引用次数: 4

2011

Comparison of SOI and Partial-SOI LDMOSFETs Using Electrical-Thermal-Stress Coupled-Field Effect

IEEE TRANSACTIONS ON ELECTRON DEVICES

Tan, Cher Ming; Huang, Guangyu

被引用次数: 3

2011

Microstructure and mechanical properties of CrN films fabricated by high power pulsed magnetron discharge plasma immersion ion implantation and deposition

APPLIED SURFACE SCIENCE

Wu, Zhongzhen; Tian, Xiubo; Wang, Zeming; Gong, Chunzhi; Yang, Shiqin; Tan, Cher Ming; Chu, Paul K.

被引用次数: 29

2011

Covalent functionalization of carbon nanotubes and their use in dielectric epoxy composites to improve heat dissipation

CARBON

Baudot, Charles; Tan, Cher Ming

被引用次数: 18

2011

Effect of hydrophilicity of carbon nanotube arrays on the release rate and activity of recombinant human bone morphogenetic protein-2

NANOTECHNOLOGY

Han, Zhao Jun; Ostrikov, Kostya (Ken); Tan, Cher Ming; Tay, Beng Kang; Peel, Sean A. F.

被引用次数: 26

2011

Modeling the effect of barrier thickness and low-k dielectric on circuit reliability using 3D model

MICROELECTRONICS RELIABILITY

He, Feifei; Tan, Cher Ming

被引用次数: 1

2010

Humidity study of a-Si PV cell

MICROELECTRONICS RELIABILITY

Tan, Cher Ming; Chen, Boon Khai Eric; Toh, Kok Peng

被引用次数: 27

2010

Circuit level interconnect reliability study using 3D circuit model

MICROELECTRONICS RELIABILITY

He, Feifei; Tan, Cher Ming

被引用次数: 23

2010

FTIR spectroscopy as a tool for nano-material characterization

INFRARED PHYSICS & TECHNOLOGY

Baudot, Charles; Tan, Cher Ming; Kong, Jeng Chien

被引用次数: 68

2010

Hot-Carrier Reliability of Power SOI EDNMOS

IEEE TRANSACTIONS ON POWER ELECTRONICS

Liao, Jie; Tan, Cher Ming; Spierings, Geert

被引用次数: 3

2010

Antibacterial action of dispersed single-walled carbon nanotubes on Escherichia coli and Bacillus subtilis investigated by atomic force microscopy

NANOSCALE

Liu, Shaobin; Ng, Andrew Keong; Xu, Rong; Wei, Jun; Tan, Cher Ming; Yang, Yanhui; Chen, Yuan

被引用次数: 152

2010

INTERFACE FRACTURE TOUGHNESS ASSESSMENT OF SOLDER JOINTS USING DOUBLE CANTILEVER BEAM TEST

INTERNATIONAL JOURNAL OF MODERN PHYSICS B

Loo, Shane Zhi Yuan; Lee, Puay Cheng; Lim, Zan Xuan; Yantara, Natalia; Tee, Tong Yan; Tan, Cher Ming; Chen, Zhong

被引用次数: 4

2010

Reply to comments on A framework to practical predictive maintenance modeling for multi-state systems

RELIABILITY ENGINEERING & SYSTEM SAFETY

Tan, Cher Ming; Raghavan, Nagarajan

被引用次数: 2	2009
Local bond average for the size and temperature dependence of elastic and vibronic properties of nanostructures	
INTERNATIONAL JOURNAL OF NANOTECHNOLOGY	
Gu, Mingxia; Sun, Chang Q.; Tan, Cher Ming; Wang, Shanzhong	
被引用次数: 2	2009
Solubility, dispersion and bonding of functionalised carbon nanotubes in epoxy resins	
INTERNATIONAL JOURNAL OF NANOTECHNOLOGY	
Baudot, Charles; Tan, Cher Ming	
被引用次数: 6	2009
SPECIAL ISSUE: ON NANOELECTRONICS	
INTERNATIONAL JOURNAL OF NANOTECHNOLOGY	
Tan, Cher Ming; Tay, Beng Kang	
被引用次数: 0	2009
Comparison of stress-induced voiding phenomena in copper line-via structures with different dielectric materials	
SEMICONDUCTOR SCIENCE AND TECHNOLOGY	
Hou, Yuejin; Tan, Cher Ming	
被引用次数: 5	2009
Analysis of humidity effects on the degradation of high-power white LEDs	
MICROELECTRONICS RELIABILITY	
Tan, Cher Ming; Chen, Boon Khai Eric; Xu, Gan; Liu, Yuanjie	
被引用次数: 40	2009
Dynamic simulation of void nucleation during electromigration in narrow integrated circuit interconnects	
JOURNAL OF APPLIED PHYSICS	
Li, Wei; Tan, Cher Ming; Raghavan, Nagarajan	
被引用次数: 14	2009
Behavior of hot carrier generation in power SOI LDNMOS with shallow trench isolation (STI)	
MICROELECTRONICS RELIABILITY	
Liao, Jie; Tan, Cher Ming; Spierings, Geert	
被引用次数: 8	2009
Electromigration in width transition copper interconnect	
MICROELECTRONICS RELIABILITY	
Roy, Arijit; Hou, Yuejin; Tan, Cher Ming	
被引用次数: 2	2009
Reverse breakdown voltage measurement for power P+NN+ rectifier	

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Huang, Guangyu; Tan, Cher Ming

被引用次数: 0

2008

Stress-induced voiding study in integrated circuit interconnects

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Hou, Yuejin; Tan, Cher Ming

被引用次数: 8

2008

Very high current density package level electromigration test for copper interconnects

JOURNAL OF APPLIED PHYSICS

Roy, Arijit; Tan, Cher Ming

被引用次数: 17

2008

A bimodal three-parameter lognormal mixture distribution for electromigration failure analysis

THIN SOLID FILMS

Tan, Cher Ming; Raghavan, Nagarajan

被引用次数: 3

2008

A framework to practical predictive maintenance modeling for multi-state systems

RELIABILITY ENGINEERING & SYSTEM SAFETY

Tan, Cher Ming; Raghavan, Nagarajan

被引用次数: 63

2008

Electromigration in ULSI interconnects

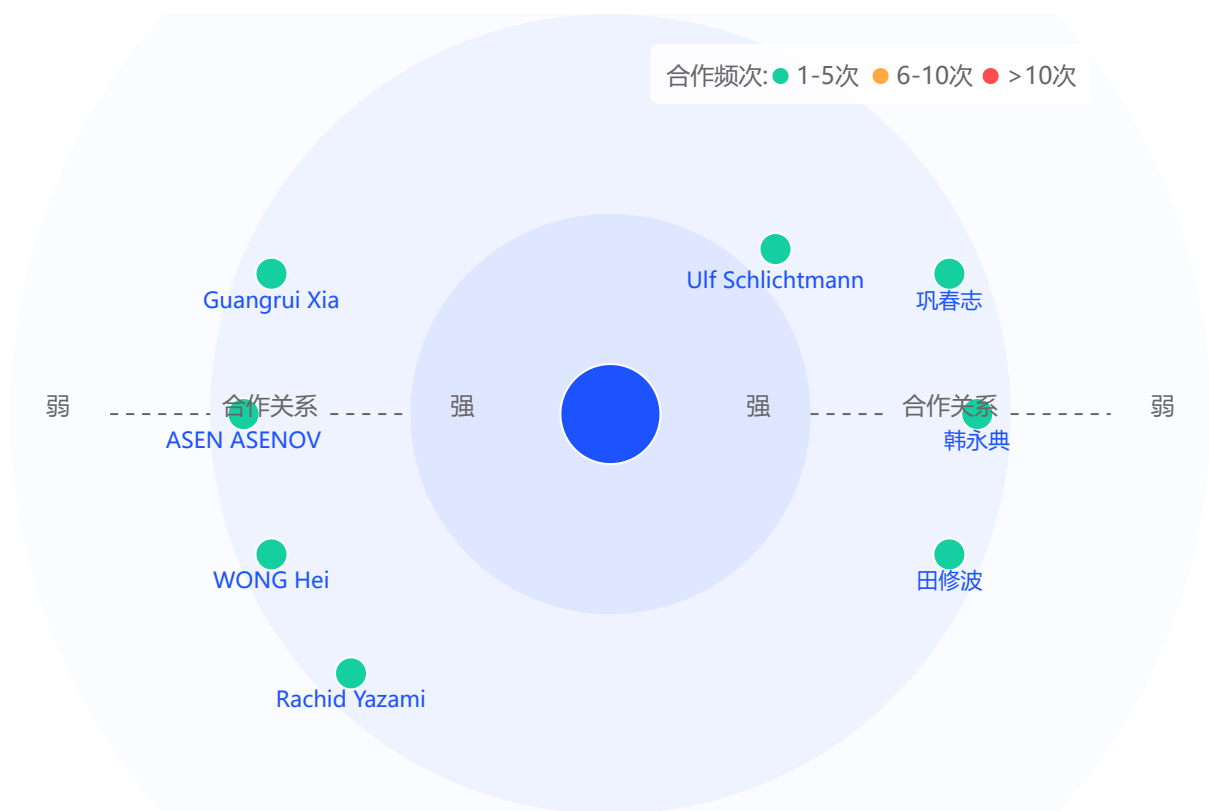
MATERIALS SCIENCE & ENGINEERING R-REPORTS

Tan, Cher Ming; Roy, Arijit

被引用次数: 194

2007

学术合作



Ulf Schlichtmann

慕尼黑工业大学

合作次数: 2

巩春志

哈尔滨工业大学

合作次数: 1

韩永典

天津大学

合作次数: 1

田修波

哈尔滨工业大学

合作次数: 1

Rachid Yazami

南洋理工大学

合作次数: 1

还有 3 个合作者